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"""Simple Stock Portfolio Tracker
Concepts: dictionary, input/output, basic
arithmetic, file handling
"""

# Hardcoded stock prices (USD per share)
STOCK_PRICES = {
    "AAPL": 180,
    "TSLA": 250,
    "GOOGL": 140,
    "MSFT": 330,
    "AMZN": 135,
}

def get_portfolio():
    """Ask the user for stock names and
    quantities. Returns {symbol: qty}."""
    portfolio = {}
    print("Available stocks:", ", ".join(f"{s}
    (${p})" for s, p in STOCK_PRICES.items()))
    print("Enter a stock symbol and quantity.
    Type 'done' when finished.\n")

    while True:
        symbol = input("Stock symbol:
        ").strip().upper()
        if symbol == "DONE":
            break
        if symbol not in STOCK_PRICES:
            print(f" '{symbol}' not found.
            Choose from: {' '.join(STOCK_PRICES)}")
            continue

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        qty_text = input(f"Quantity of {symbol}:
").strip()
        try:
            qty = int(qty_text)
            if qty <= 0:
                raise ValueError
        except ValueError:
            print(" Please enter a positive
whole number.")
            continue

        portfolio[symbol] =
portfolio.get(symbol, 0) + qty
        return portfolio

def calculate_total(portfolio):
    """Return (list of rows, total). Each row =
(symbol, qty, price, value)."""
    rows = []
    total = 0
    for symbol, qty in portfolio.items():
        price = STOCK_PRICES[symbol]
        value = qty * price
        rows.append((symbol, qty, price, value))
        total += value
    return rows, total

def display_summary(rows, total):
    print("\n" + "=" * 42)
    print(f"{'Stock':<8}{'Qty':>6}{'Price':>10}
{'Value':>14}")
    print("-" * 42)
    for symbol, qty, price, value in rows:
        print(f"{symbol:<8}{qty:>6}
{price:>10,.2f}{value:>14,.2f}")

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        print("-" * 42)
        print(f"'TOTAL INVESTMENT':<24}
{total:>18,.2f}")
        print("=" * 42)

def save_to_file(rows, total):
    choice = input("\nSave result to a file?
(txt/csv/no): ").strip().lower()
    if choice == "txt":
        with open("portfolio.txt", "w") as f:
            f.write("Stock Portfolio Summary\n")
            f.write("-" * 30 + "\n")
            for symbol, qty, price, value in
rows:
                f.write(f"{symbol}: {qty} x
${price} = ${value:,.2f}\n")
            f.write("-" * 30 + "\n")
            f.write(f"Total Investment:
${total:,.2f}\n")
            print("Saved to portfolio.txt")
    elif choice == "csv":
        with open("portfolio.csv", "w") as f:
f.write("Stock,Quantity,Price,Value\n")
            for symbol, qty, price, value in
rows:
                f.write(f"{symbol},{qty},
{price},{value}\n")
            f.write(f"TOTAL,,,{total}\n")
            print("Saved to portfolio.csv")
    else:
        print("Not saved.")

def main():
    portfolio = get_portfolio()

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    if not portfolio:
        print("No stocks entered. Goodbye!")
        return
    rows, total = calculate_total(portfolio)
    display_summary(rows, total)
    save_to_file(rows, total)

if __name__ == "__main__":
    main()
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